

Summer Bridge Practice Packet

Stay sharp over the summer and walk into 6th grade ready. A few problems a day keeps your Grade 5 skills fresh while you preview the big ideas of Grade 6 math.

Name: _____

Teacher: _____

Start date: _____ Finish date: _____

What this packet is

This packet reviews the most important math from Grade 5 and gives you a head start on Grade 6. Each Part begins with vocabulary and a worked example, then gives you practice sets, a **Level 1** set (extra support, friendlier numbers) and a **Level 2** set (a stretch), and a Quick Check you can grade yourself. Answer keys for every Part are in the back.

How to use this packet

Plan for about **4–6 weeks**. Aim for **5–8 problems a day**, 4–5 days a week — short and steady beats one long cram. Try this routine:

- Read the vocabulary and the Worked Example *first*.
- Do the regular practice, then try the Level 2 stretch if you feel ready.
- Use the Quick Check to see how you did, then look at the answer key.
- Stuck? Tap the reference chips at the top of each Part to relearn it on the site.
- Check off each Part as you finish: ☐ Part 1 ☐ Part 2 ☐ Part 3 ☐ Part 4 ☐ Part 5 ☐ Part 6 ☐ Spiral Review

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7. Cumulative / Spiral Review · mixed
8. Answer Keys · for teachers & parents

Standards & Skills Map

Part	Skill focus	Standards	Where to review
1	Place value, compare/round, expanded form, decimal +, −, ×, ÷, estimation	5.NBT	Unit 1 Notes
2	Equivalent & simplest form, compare, +/− unlike denominators, × and ÷ fractions, mixed numbers	5.NF	Unit 2 Notes
3	Multi-digit × and ÷ with remainders, GCF, LCM, prime/composite, divisibility	5.NBT, 6.NS.B	Unit 1 Notes
4	Ratio language, equivalent ratios & tables, unit rate, simple percents, fraction–decimal–percent	6.RP.A	Unit 3 · Unit 4
5	Write/evaluate expressions, order of operations, input–output patterns, plotting points	5.OA, 6.EE, 6.NS.C	Unit 6 · Unit 9
6	Area, perimeter, volume of prisms, unit conversion, mean/median/mode/range, line plots	5.MD, 5.G, 6.SP, 6.G	Unit 5 · Unit 10

"Where to review" — how the chips work

Each Part header has small rounded **reference chips**. Each chip is a link to real notes, slides, or practice on the Neft Teacher site. If a skill feels rusty, tap a chip to relearn it *before* you practice. For example, [Unit 3 Notes](#) opens the ratio lesson, [Vocabulary](#) opens the vocab deck, and [Spiral Review](#) opens extra mixed practice.

1

Place Value, Whole Numbers & Decimal Operations

5.NBT

REVIEW:

[Unit 1 Notes \(Number Sense\)](#)

[Get Ready](#)

[Vocabulary](#)

VOCABULARY

Place value

The value a digit has because of its position (e.g., the 3 in 4.3 means 3 tenths).

Expanded form

A number written as the sum of each digit's value, like $305 = 300 + 5$.

Round

Replace a number with a nearby simpler value to a given place.

Decimal point

The dot that separates whole numbers from parts of a whole.

Estimate

A close, reasonable answer found using friendly numbers.

Product / Quotient

Product is the answer to multiplication; quotient is the answer to division.

WATCH ONE FIRST

Multiply decimals: 3.4×1.6

Ignore the dots: $34 \times 16 = 544$.

Count decimal places in the problem: $3.\underline{4}$ has 1, $1.\underline{6}$ has 1, so 2 total.

Put 2 decimal places in the answer: $544 \rightarrow \mathbf{5.44}$.

Divide decimals: $7.2 \div 0.8$

Move both decimals to make the divisor whole: $0.8 \rightarrow 8$, so $7.2 \rightarrow 72$.

Now divide: $72 \div 8 = \mathbf{9}$.

1A. Compare, round & expanded form Use $<$, $>$, or $=$.

1. Compare: 0.7 ____ 0.70
2. Compare: 3.45 ____ 3.5
3. Compare: 12.08 ____ 12.1
4. Compare: 6.300 ____ 6.3
5. Round 4.68 to the nearest tenth.
6. Round 27.349 to the nearest hundredth.
7. Round 8.95 to the nearest tenth.
8. Write 7.42 in expanded form (use place values).
9. Write $500 + 30 + 0.6$ as a single number.
10. Order from least to greatest: 2.5, 2.05, 2.55, 2.50.

1B. Add & subtract decimals Line up the decimal points.

1. $3.6 + 2.8$
2. $12.45 + 7.9$
3. $9.5 - 3.7$
4. $20 - 6.35$
5. $0.45 + 0.6 + 1.2$
6. $15.04 - 8.6$
7. $\$24.50 + \13.75
8. $\$50.00 - \37.85

1C. Multiply & divide decimals

1. 2.5×4
2. 1.2×0.3
3. 6.4×1.5
4. 0.25×8
5. $4.8 \div 4$
6. $9.6 \div 0.6$
7. $7.5 \div 0.5$
8. $3.0 \div 8$

1D. Estimation & find-the-error Find and fix the mistake.

1. Estimate $4.9 + 6.1$ by rounding each to the nearest whole number.
2. Estimate 19.8×4 by rounding 19.8 to the nearest whole number.
3. A student says $0.3 \times 0.2 = 0.6$. Find and fix the error.
4. A student says $5.6 + 4 = 6.0$. Find and fix the error.

LEVEL 1 **Extra support** Friendlier numbers.

1. Compare: 0.4 ____ 0.6
2. Round 3.7 to the nearest whole number.
3. $2.1 + 1.5$
4. $4.8 - 1.3$
5. 2×0.5
6. $6 \div 0.6$

LEVEL 2 **Stretch**

1. Lin buys 3 notebooks at \$2.45 each and pays with a \$10 bill. How much change does she get?
2. A ribbon is 4.5 m long. It is cut into 0.9 m pieces. How many pieces are there?
3. Round 6.748 to the nearest hundredth, then to the nearest tenth. Are the answers the same?
4. $0.125 \times 8 = ?$ Explain why the answer is a whole number.

QUICK CHECK

1. Compare: 5.09 ____ 5.9
2. $7.4 - 2.85$
3. 1.5×0.4
4. $8.1 \div 0.9$

REVIEW:

Unit 2 Notes (Fraction Division)

Unit 1 Notes

Vocabulary

VOCABULARY

Numerator / Denominator

Top number (parts you have) / bottom number (total equal parts).

Equivalent fractions

Different fractions that name the same amount, like $1/2 = 2/4$.

Simplest form

A fraction with no common factor left to divide out, like $6/8 = 3/4$.

Common denominator

A shared bottom number so fractions can be added or compared.

Mixed number

A whole number plus a fraction, like $2 \frac{1}{3}$.

Reciprocal

A fraction flipped over; you multiply by it to divide ($3/4 \rightarrow 4/3$).

WATCH ONE FIRST

Add unlike denominators: $1/3 + 1/4$

Common denominator of 3 and 4 is 12.

Rename: $1/3 = 4/12$ and $1/4 = 3/12$.

Add the numerators: $4/12 + 3/12 = 7/12$.

Divide fractions: $3/4 \div 1/2$

Keep, flip, multiply: $3/4 \times 2/1$.

$3 \times 2 = 6$, $4 \times 1 = 4$, so $6/4 = 1 \frac{1}{2}$.

2A. Equivalent, simplify & compare Use $<$, $>$, or $=$.1. Fill in: $2/3 = ?/12$ 3. Simplify $8/12$.2. Fill in: $3/5 = ?/20$ 4. Simplify $9/15$.

5. Simplify $10/4$ to a mixed number in simplest form.

6. Compare: $2/3$ ____ $3/4$

7. Compare: $5/6$ ____ $7/8$

8. Compare: $4/10$ ____ $2/5$

2B. Add & subtract fractions

Find a common denominator. Simplify answers.

1. $1/2 + 1/3$

2. $3/4 + 1/8$

3. $5/6 - 1/3$

4. $7/10 - 1/5$

5. $2/3 + 3/4$

6. $1\ 1/2 + 2\ 1/4$

7. $3\ 1/3 - 1\ 1/6$

8. $4 - 1\ 2/3$

2C. Multiply & divide fractions

1. $1/2 \times 3/5$

2. $2/3 \times 3/4$

3. $4/5 \times 10$

4. $2\ 1/2 \times 1\ 1/3$

5. $1/2 \div 1/4$

6. $3/5 \div 2/5$

7. $6 \div 1/3$

8. $2\ 1/4 \div 3/4$

2D. Word problems & find-the-error

1. A recipe needs $3/4$ cup of sugar. You make a half batch. How much sugar?

2. Maya runs $2/3$ mile and Tom runs $5/6$ mile. Who runs farther, and by how much?

3. A 5-foot board is cut into $2/3$ -foot pieces. How many pieces?

4. A student says $1/2 + 1/3 = 2/5$. Find and fix the error.

LEVEL 1 Extra support

Like denominators or halves.

1. $1/4 + 2/4$

2. $3/5 - 1/5$

3. $1/2 \times 1/2$

4. Simplify $4/8$.

5. $2 \div 1/2$

6. Compare: $1/2$ ____ $1/4$

LEVEL 2 Stretch

1. Jordan ate $1/4$ of a pizza and his friend ate $3/8$. How much is left?

2. How many $3/8$ -cup scoops fill a 3-cup container?

3. $3/4$ of a class is 18 students. How many students are in the whole class?

4. $1\ 1/2 \times 2/3 \div 1/2 = ?$ (work left to right)

QUICK CHECK

1. Simplify $12/16$.
2. $2/3 + 1/6$
3. $3/4 \times 8$
4. $3/4 \div 1/4$

Multiplication/Division Fluency, Factors & Multiples

5.NBT · 6.NS.B

REVIEW:

Unit 1 Notes

Get Ready

Spiral Review

VOCABULARY

Factor

A whole number that divides evenly into another (4 is a factor of 12).

Multiple

The result of multiplying by whole numbers (12, 24, 36 are multiples of 12).

GCF

Greatest Common Factor: the largest factor two numbers share.

LCM

Least Common Multiple: the smallest multiple two numbers share.

Prime / Composite

Prime has exactly two factors (1 and itself); composite has more.

Divisible

Divides evenly with no remainder (24 is divisible by 6).

WATCH ONE FIRST

Long division with remainder: $487 \div 6$

6 goes into 48 eight times (48). Bring down 7 $\rightarrow$ 7.

6 goes into 7 once (6), remainder 1.

Answer: **81 R1** (check: $81 \times 6 + 1 = 487$).

GCF and LCM of 8 and 12:

Factors of 8: 1, 2, 4, 8. Factors of 12: 1, 2, 3, 4, 6, 12. Largest shared = **GCF 4**.

Multiples of 8: 8, 16, 24... Multiples of 12: 12, 24... Smallest shared = **LCM 24**.

3A. Multi-digit multiplication

- | | |
|-------------------|-------------------|
| 1. 34×6 | 4. 56×23 |
| 2. 27×15 | 5. 208×7 |
| 3. 123×4 | 6. 45×30 |

3B. Division (some with remainders)

- | | |
|-----------------|------------------|
| 1. $144 \div 6$ | 4. $365 \div 8$ |
| 2. $250 \div 5$ | 5. $936 \div 12$ |
| 3. $487 \div 4$ | 6. $500 \div 7$ |

3C. Factors, GCF & LCM

- | | |
|----------------------------|---------------------|
| 1. List all factors of 24. | 5. LCM of 4 and 6 |
| 2. List all factors of 18. | 6. LCM of 3 and 5 |
| 3. GCF of 12 and 18 | 7. GCF of 20 and 30 |
| 4. GCF of 16 and 24 | 8. LCM of 6 and 8 |

3D. Prime/composite, divisibility & find-the-error

- | | |
|---|---|
| 1. Is 29 prime or composite? | 5. A student says the GCF of 6 and 9 is 18. Find and fix the error. |
| 2. Is 51 prime or composite? Explain. | |
| 3. Is 246 divisible by 3? (Hint: digit sum) | 6. A student says 1 is prime. Find and fix the error. |
| 4. Is 135 divisible by 5? by 2? | |

LEVEL 1 Extra support

- | | |
|----------------------------|-----------------------------|
| 1. 23×3 | 4. GCF of 6 and 8 |
| 2. $48 \div 4$ | 5. LCM of 2 and 5 |
| 3. List the factors of 10. | 6. Is 7 prime or composite? |

LEVEL 2 Stretch

- | | |
|---|--|
| 1. Hot dogs come in packs of 8 and buns in packs of 12. What is the fewest of each pack to have equal numbers? (Use LCM.) | 2. You have 24 red and 36 blue beads to split into identical bracelets with none left over. What is the greatest number of bracelets? (Use GCF.) |
|---|--|

3. A bus holds 14 students. How many buses are needed for 100 students?

4. Write 36 as a product of prime factors.

QUICK CHECK

1. 38×14

2. $425 \div 6$

3. GCF of 24 and 36

4. LCM of 4 and 10

REVIEW:

Unit 3 Notes (Ratios)

Unit 4 (Rates & Percents)

Vocabulary

VOCABULARY

Ratio

A comparison of two quantities, written 3:4 or 3 to 4.

Equivalent ratios

Ratios that show the same relationship, like $2:3 = 4:6$.

Rate

A ratio comparing two different units, like miles per hour.

Unit rate

A rate "per 1," like 60 miles per 1 hour.

Percent

A ratio out of 100; 25% means 25 per 100.

Equivalent forms

The same value as a fraction, decimal, and percent ($1/2 = 0.5 = 50\%$).

WATCH ONE FIRST

Unit rate: 12 apples cost \$6. Find the cost per apple.

Set up the rate: \$6 for 12 apples.

Divide to get "per 1": $6 \div 12 = \text{\$0.50 per apple}$.

Percent of a number: 25% of 80

$25\% = 25/100 = 1/4$.

$1/4$ of 80 = $80 \div 4 = 20$.

4A. Ratio language & equivalent ratios

1. A box has 3 red and 5 blue marbles.
Write the ratio of red to blue.
2. Write the ratio of blue to total for the box above.
3. Complete: $2:3 = 8:?$
4. Complete: $4:5 = ?:20$
5. Are 6:9 and 2:3 equivalent? Explain.
6. Fill the table for $1:4 \rightarrow (2,?), (3,?), (5,?)$.

4B. Rates & unit rates

1. A car goes 150 miles in 3 hours. Unit rate (mph)?
2. 8 oranges cost \$4. Cost per orange?
3. A printer prints 90 pages in 5 minutes. Pages per minute?
4. Which is the better buy: 6 pens for \$3 or 10 pens for \$4?
5. A runner goes 400 m in 80 seconds. Meters per second?
6. \$24 for 6 tickets. Price per ticket?

4C. Percents & out-of-100

1. Write 37 out of 100 as a percent.
2. 50% of 60
3. 10% of 90
4. 25% of 40
5. 20% of 50
6. 75% of 80

4D. Fraction–decimal–percent & find-the-error

1. Write $\frac{1}{2}$ as a decimal and a percent.
2. Write 0.25 as a fraction and a percent.
3. Write $\frac{3}{4}$ as a percent.
4. Write 0.6 as a percent.
5. A student says $\frac{1}{4} = 25\%$ so $\frac{1}{4} = 0.25...$ but then writes $\frac{1}{4} = 4\%$. Find and fix the error.
6. A student says 30% of 200 is 6. Find and fix the error.

LEVEL 1 Extra support

1. Write the ratio of 2 cats to 3 dogs.
2. Complete: $1:2 = 4:?$
3. 6 candies cost \$3. Cost per candy?
4. 50% of 20
5. 10% of 40
6. Write $\frac{1}{2}$ as a percent.

LEVEL 2 Stretch

1. A recipe uses 2 cups flour to 3 cups water. How much water for 8 cups flour?
2. A store has 30% off a \$50 jacket. What is the sale price?
3. A class is 60% girls. If there are 30 students, how many are boys?
4. Which is cheaper per ounce: 12 oz for \$3.60 or 16 oz for \$4.00?

QUICK CHECK

1. Complete: $3:4 = ? : 16$
2. 120 miles in 2 hours. Unit rate (mph)?
3. 20% of 150
4. Write $\frac{3}{5}$ as a percent.

REVIEW:

Unit 6 Notes (Expressions)

Unit 9 (Coordinate Plane)

Spiral Review

VOCABULARY

Expression

A math phrase with numbers and operations, like $3 + 4 \times 2$ (no equals sign).

Variable

A letter that stands for a number, like x .

Evaluate

Find the value of an expression by plugging in numbers.

Order of operations

Do grouping, then $\times$ and $\div$ left to right, then $+$ and $-$ left to right.

Coordinate plane

A grid with an x -axis and a y -axis used to plot points.

Ordered pair

(x, y) : how far right/left, then up/down, from the origin $(0, 0)$.

WATCH ONE FIRST

Order of operations: $3 + 4 \times 2$

Multiply first: $4 \times 2 = 8$.

Then add: $3 + 8 = 11$.

Evaluate: $2x + 5$ when $x = 6$

Replace x : $2(6) + 5$.

$12 + 5 = 17$.

Plot: The point $(3, 2)$ is 3 right and 2 up from the origin (Quadrant I).

5A. Order of operations

1. $8 + 2 \times 3$

2. $(8 + 2) \times 3$

3. $20 - 6 \div 2$

4. $4 \times (5 - 2)$

5. $3 \times 2 + 4 \times 5$

6. $18 \div (1 + 2) + 7$

5B. Write & evaluate expressions

1. Write an expression: "5 more than a number x ."
2. Write an expression: "twice a number n ."
3. Write an expression: "the cost of t tickets at \$9 each."

4. Evaluate $x + 7$ when $x = 12$.
5. Evaluate $3n$ when $n = 8$.
6. Evaluate $4x - 1$ when $x = 5$.
7. Evaluate $2(a + 3)$ when $a = 4$.
8. Evaluate $30 \div y$ when $y = 6$.

5C. Input–output patterns

1. Rule: $\text{out} = \text{in} + 4$. Find out when $\text{in} = 7$.
2. Rule: $\text{out} = 3 \times \text{in}$. Find out when $\text{in} = 5$.
3. Rule: $\text{out} = 2 \times \text{in} + 1$. Find out when $\text{in} = 6$.
4. A table shows in: 1, 2, 3 and out: 5, 10, 15. Write the rule.
5. A table shows in: 2, 4, 6 and out: 5, 7, 9. Write the rule.
6. Continue the pattern: 3, 6, 12, 24, ____.

5D. Coordinate plane & find-the-error

1. In which quadrant is $(4, 5)$?
2. In which quadrant is $(-3, 2)$?
3. In which quadrant is $(-2, -6)$?
4. What are the coordinates of the origin?
5. Point A is at $(5, 0)$. Is it on the x -axis or y -axis?
6. A student plots $(3, 2)$ by going up 3 then right 2. Find and fix the error.

LEVEL 1 Extra support

1. $5 + 3 \times 2$
2. $(6 + 4) \div 2$
3. Evaluate $x + 5$ when $x = 3$.
4. Evaluate $2n$ when $n = 4$.
5. Rule: $\text{out} = \text{in} + 2$. Find out when $\text{in} = 9$.
6. In which quadrant is $(1, 1)$?

LEVEL 2 Stretch

1. Evaluate $2(x + 3) - 4$ when $x = 5$.
2. A taxi charges \$3 plus \$2 per mile. Write an expression for m miles, then find the cost of 7 miles.
3. Plot $(2, 3)$, $(2, -1)$, and $(-2, 3)$. Which two share the same x -value?
4. A table: $\text{in } 1 \rightarrow 4, 2 \rightarrow 7, 3 \rightarrow 10$. Write the rule, then find out when $\text{in} = 10$.

QUICK CHECK

1. $12 - 2 \times 3$
2. Evaluate $4x + 2$ when $x = 5$.
3. Rule: $\text{out} = 2 \times \text{in} + 3$. Find out when $\text{in} = 4$.
4. In which quadrant is $(-4, -1)$?

REVIEW:

Unit 5 Notes (Area)

Unit 10 (Volume)

Unit 8 (Statistics)

VOCABULARY**Area**

The space inside a flat shape, measured in square units.

Perimeter

The distance all the way around a shape.

Volume

The space inside a solid, measured in cubic units; box volume = $l \times w \times h$.

Mean

The average: add the values, then divide by how many there are.

Median / Mode

Median is the middle value in order; mode is the most frequent value.

Range

The largest value minus the smallest value.

WATCH ONE FIRST

Area of a triangle: base 8, height 5

Area = $\frac{1}{2} \times \text{base} \times \text{height}$.

$\frac{1}{2} \times 8 \times 5 = \frac{1}{2} \times 40 = \mathbf{20 \text{ square units.}}$

Mean: data 4, 6, 8, 10

Add: $4 + 6 + 8 + 10 = 28$.

Divide by 4: $28 \div 4 = \mathbf{7}$.

Volume of a box: $3 \times 4 \times 2 = \mathbf{24 \text{ cubic units.}}$

6A. Area & perimeter

1. Area of a rectangle 7 by 4.

2. Perimeter of a rectangle 7 by 4.

- Area of a square with side 6.
- Area of a triangle: base 10, height 6.
- Area of a triangle: base 9, height 4.
- Composite: an L-shape made of a 5×3 rectangle and a 2×2 rectangle. Total area?

6B. Volume & unit conversion

- Volume of a box $5 \times 3 \times 2$.
- Volume of a cube with edge 4.
- Volume of a box $6 \times 2 \times 5$.
- Convert 3 feet to inches (1 ft = 12 in).
- Convert 2,000 g to kilograms (1 kg = 1,000 g).
- Convert 4 m to centimeters (1 m = 100 cm).

6C. Data: mean, median, mode, range Data set: 3, 7, 7, 8, 5.

- Find the mean of 3, 7, 7, 8, 5.
- Find the median of 3, 7, 7, 8, 5.
- Find the mode of 3, 7, 7, 8, 5.
- Find the range of 3, 7, 7, 8, 5.
- Find the mean of 10, 12, 14, 16.
- Find the median of 2, 4, 6, 8, 10, 12.

6D. Line plots & find-the-error

- On a line plot, 4 X's sit above "2", and 1 X above "5". How many data points in all?
- For the plot above, what value appears most often?
- A line plot shows lengths $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$. What is the total of all four lengths?
- A student finds the mean of 2, 4, 6 by adding to get 12, then says the mean is 6. Find and fix the error.
- A student says the range of 5, 9, 9, 2 is 9. Find and fix the error.
- A student says volume of a $2 \times 3 \times 4$ box is 9 (added the sides). Find and fix the error.

LEVEL 1 Extra support

- Area of a rectangle 5 by 2.
- Perimeter of a square with side 3.
- Area of a triangle: base 4, height 6.
- Volume of a box $2 \times 2 \times 2$.
- Mean of 4, 6, 8.
- Range of 3, 8, 5.

LEVEL 2 Stretch

1. A garden is 12 ft by 8 ft. A 1-ft border path goes around the inside edge. What is the area of the inner rectangle (10 ft by 6 ft)?
2. A fish tank is 20 in $\times$ 10 in $\times$ 12 in. What is its volume?
3. Five test scores are 80, 85, 90, 95, and one unknown. The mean is 88. Find the missing score.
4. The median of 4, 7, x , 12 (already in order) is 9. Find x .

QUICK CHECK

1. Area of a triangle: base 12, height 5.
2. Volume of a box $4 \times 3 \times 3$.
3. Mean of 6, 8, 10, 12.
4. Range of 14, 9, 20, 11.



Cumulative / Spiral Review

MIXED

REVIEW:

Spiral Review

MCAP Review

Unit Map

Spiral Set A A little of everything.

1. Round 5.36 to the nearest tenth.
2. $4.7 + 3.85$
3. 2.5×0.4
4. Simplify $9/12$.
5. $1/2 + 1/3$
6. $3/4 \times 8$
7. GCF of 18 and 24
8. LCM of 4 and 6
9. Complete: $3:5 = 9:?$
10. 25% of 60
11. Evaluate $3x + 2$ when $x = 4$.
12. $8 + 2 \times 5$
13. Area of a rectangle 6 by 5.
14. Mean of 5, 7, 9.

Spiral Set B A little of everything.

1. Compare: 0.6 ____ 0.59
2. $10 - 4.25$
3. $6.3 \div 0.7$
4. $2/3 - 1/6$
5. $1\frac{1}{2} + 2\frac{1}{4}$
6. $3/4 \div 1/2$
7. $365 \div 5$
8. Is 39 prime or composite?
9. A car goes 180 miles in 3 hours. Unit rate (mph)?
10. Write 0.75 as a percent.
11. Evaluate $2(n + 1)$ when $n = 5$.
12. In which quadrant is $(-2, 3)$?
13. Volume of a box $4 \times 2 \times 3$.
14. Median of 3, 6, 9, 12, 15.



Answer Keys

FOR TEACHERS & PARENTS

Compact keys for every Part, the Level sets, the Quick Checks, and the Spiral Review. Some open-ended explanations may vary; sample answers shown.

Part 1 — Place Value & Decimal Operations

1A.

- | | | |
|------|---------------------|----------------------------------|
| 1. = | 5. 4.7 | 9. 530.6 |
| 2. < | 6. 27.35 | 10. 2.05, 2.5, 2.50, 2.55 (2.5 = |
| 3. < | 7. 9.0 (or 9) | 2.50) |
| 4. = | 8. $7 + 0.4 + 0.02$ | |

1B.

- | | | |
|----------|----------|------------|
| 1. 6.4 | 4. 13.65 | 7. \$38.25 |
| 2. 20.35 | 5. 2.25 | 8. \$12.15 |
| 3. 5.8 | 6. 6.44 | |

1C.

- | | | |
|---------|--------|----------|
| 1. 10 | 4. 2 | 7. 15 |
| 2. 0.36 | 5. 1.2 | 8. 0.375 |
| 3. 9.6 | 6. 16 | |

1D.

- | | | |
|-----------------------|--------------------------------------|---------------------------|
| 1. $5 + 6 = 11$ | 3. Wrong; $0.3 \times 0.2 = 0.06$ (2 | 4. Wrong; $5.6 + 4 = 9.6$ |
| 2. $20 \times 4 = 80$ | decimal places) | |

Level 1.

- | | | |
|------|--------|-------|
| 1. < | 3. 3.6 | 5. 1 |
| 2. 4 | 4. 3.5 | 6. 10 |

Level 2.

- | | |
|------------------------------|--|
| 1. $\$10 - \$7.35 = \$2.65$ | 3. 6.75 and 6.7; no, different |
| 2. $4.5 \div 0.9 = 5$ pieces | 4. 1; $0.125 = 1/8$, and $1/8 \times 8 = 1$ |

Quick Check.

- | | |
|---------|--------|
| 1. < | 3. 0.6 |
| 2. 4.55 | 4. 9 |

Part 2 — Fractions

2A.

1. 8

2. 12

3. $\frac{2}{3}$

4. $\frac{3}{5}$

5. $2\frac{1}{2}$

6. $<$

7. $<$

8. $=$

2B.

1. $\frac{5}{6}$

2. $\frac{7}{8}$

3. $\frac{1}{2}$

4. $\frac{1}{2}$

5. $1\frac{5}{12}$

6. $3\frac{3}{4}$

7. $2\frac{1}{6}$

8. $2\frac{1}{3}$

2C.

1. $\frac{3}{10}$

2. $\frac{1}{2}$

3. 8

4. $3\frac{1}{3}$

5. 2

6. $1\frac{1}{2}$

7. 18

8. 3

2D.

1. $\frac{3}{8}$ cup

2. Tom, by $\frac{1}{6}$ mile

3. 7 pieces (with $\frac{1}{3}$ ft left over)

4. Wrong; need common denom:
 $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$

Level 1.

1. $\frac{3}{4}$

2. $\frac{2}{5}$

3. $\frac{1}{4}$

4. $\frac{1}{2}$

5. 4

6. $>$

Level 2.

1. $\frac{3}{8}$ left

2. 8 scoops

3. 24 students

4. 2

Quick Check.

1. $\frac{3}{4}$

2. $\frac{5}{6}$

3. 6

4. 3

Part 3 — Fluency, Factors & Multiples

3A.

- | | | |
|--------|----------|----------|
| 1. 204 | 3. 492 | 5. 1,456 |
| 2. 405 | 4. 1,288 | 6. 1,350 |

3B.

- | | | |
|-------|-----------|----------|
| 1. 24 | 3. 121 R3 | 5. 78 |
| 2. 50 | 4. 45 R5 | 6. 71 R3 |

3C.

- | | | |
|-----------------------------|-------|-------|
| 1. 1, 2, 3, 4, 6, 8, 12, 24 | 4. 8 | 7. 10 |
| 2. 1, 2, 3, 6, 9, 18 | 5. 12 | 8. 24 |
| 3. 6 | 6. 15 | |

3D.

- | | | |
|-------------------------------------|---|---|
| 1. Prime | 3. Yes ($2+4+6 = 12$, divisible by 3) | 5. Wrong; GCF of 6 and 9 is 3 (18 is the LCM) |
| 2. Composite ($51 = 3 \times 17$) | 4. By 5 yes; by 2 no | 6. Wrong; 1 is neither prime nor composite |

Level 1.

- | | | |
|-------|----------------|----------|
| 1. 69 | 3. 1, 2, 5, 10 | 5. 10 |
| 2. 12 | 4. 2 | 6. Prime |

Level 2.

- | | | |
|---|---------------------------------|-----------------------------------|
| 1. 3 packs of hot dogs, 2 packs of buns (24 each) | 2. 12 bracelets (GCF of 24, 36) | 4. $2 \times 2 \times 3 \times 3$ |
| | 3. 8 buses | |

Quick Check.

- | | |
|----------|-------|
| 1. 532 | 3. 12 |
| 2. 70 R5 | 4. 20 |

Part 4 — Ratios, Rates & Percents

4A.

1. 3:5
2. 5:8

3. 12
4. 16

5. Yes; $6:9 \div 3 = 2:3$
6. (2,8), (3,12), (5,20)

4B.

1. 50 mph
2. \$0.50 per orange

3. 18 pages/min
4. 10 for \$4 (\$0.40 vs \$0.50 each)

5. 5 m/s
6. \$4 per ticket

4C.

1. 37%
2. 30

3. 9
4. 10

5. 10
6. 60

4D.

1. 0.5 and 50%
2. $\frac{1}{4}$ and 25%

3. 75%
4. 60%

5. Wrong; $\frac{1}{4} = 0.25 = 25\%$, not 4%
6. Wrong; 30% of 200 = 60

Level 1.

1. 2:3
2. 8

3. \$0.50 per candy
4. 10

5. 4
6. 50%

Level 2.

1. 12 cups water
2. \$35

3. 12 boys (40% of 30)
4. 16 oz (\$0.25/oz vs \$0.30/oz)

Quick Check.

1. 12
2. 60 mph

3. 30
4. 60%

Part 5 — Expressions, Patterns & Coordinate Plane

5A.

1. 14

2. 30

3. 17

4. 12

5. 26

6. 13

5B.

1. $x + 5$

2. $2n$

3. $9t$

4. 19

5. 24

6. 19

7. 14

8. 5

5C.

1. 11

2. 15

3. 13

4. $\text{out} = 5 \times \text{in}$

5. $\text{out} = \text{in} + 3$

6. 48

5D.

1. Quadrant I

2. Quadrant II

3. Quadrant III

4. (0, 0)

5. x-axis

6. Wrong; go right 3 then up 2 (x first, then y)

Level 1.

1. 11

2. 5

3. 8

4. 8

5. 11

6. Quadrant I

Level 2.

1. 12

2. $3 + 2m$; \$17

3. (2,3) and (2,-1) share $x = 2$

4. $\text{out} = 3 \times \text{in} + 1$; 31

Quick Check.

1. 6

2. 22

3. 11

4. Quadrant III

Part 6 — Geometry, Measurement & Data

6A.

- | | | |
|----------------|----------------|-----------------------------|
| 1. 28 sq units | 3. 36 sq units | 5. 18 sq units |
| 2. 22 units | 4. 30 sq units | 6. 19 sq units ($15 + 4$) |

6B.

- | | | |
|-------------------|-------------------|-----------|
| 1. 30 cubic units | 3. 60 cubic units | 5. 2 kg |
| 2. 64 cubic units | 4. 36 in | 6. 400 cm |

6C.

- | | | |
|----------------------|------|------------------------|
| 1. 6 ($30 \div 5$) | 3. 7 | 5. 13 |
| 2. 7 | 4. 5 | 6. 7 (mean of 6 and 8) |

6D.

- | | | |
|------------------|---------------------------------------|--|
| 1. 5 data points | 3. $1 \frac{3}{4}$ | 5. Wrong; range = $9 - 2 = 7$ |
| 2. 2 | 4. Wrong; $12 \div 3 = 4$, mean is 4 | 6. Wrong; multiply: $2 \times 3 \times 4 = 24$ |

Level 1.

- | | | |
|----------------|------------------|------|
| 1. 10 sq units | 3. 12 sq units | 5. 6 |
| 2. 12 units | 4. 8 cubic units | 6. 5 |

Level 2.

- | | | |
|-------------------|--|--|
| 1. 60 sq ft | 3. 90 (sum must be 440; $440 - 350 = 90$) | 4. 11 (median of 7 and x is 9, so $x = 11$) |
| 2. 2,400 cubic in | | |

Quick Check.

- | | |
|-------------------|-------|
| 1. 30 sq units | 3. 9 |
| 2. 36 cubic units | 4. 11 |

Spiral Review — Set A

- | | | |
|------------------|--------|-----------------|
| 1. 5.4 | 6. 6 | 11. 14 |
| 2. 8.55 | 7. 6 | 12. 18 |
| 3. 1.0 (or 1) | 8. 12 | 13. 30 sq units |
| 4. $\frac{3}{4}$ | 9. 15 | 14. 7 |
| 5. $\frac{5}{6}$ | 10. 15 | |

Spiral Review — Set B

- | | | |
|-------------------|-------------------------------------|--------------------|
| 1. > | 6. $1\frac{1}{2}$ | 11. 12 |
| 2. 5.75 | 7. 73 | 12. Quadrant II |
| 3. 9 | 8. Composite ($39 = 3 \times 13$) | 13. 24 cubic units |
| 4. $\frac{1}{2}$ | 9. 60 mph | 14. 9 |
| 5. $3\frac{3}{4}$ | 10. 75% | |